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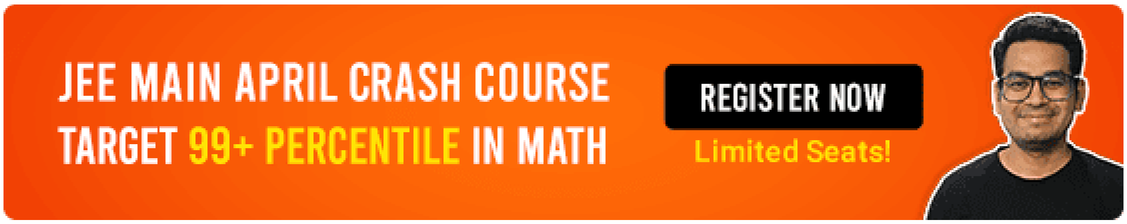
For JEE Main 2020 April



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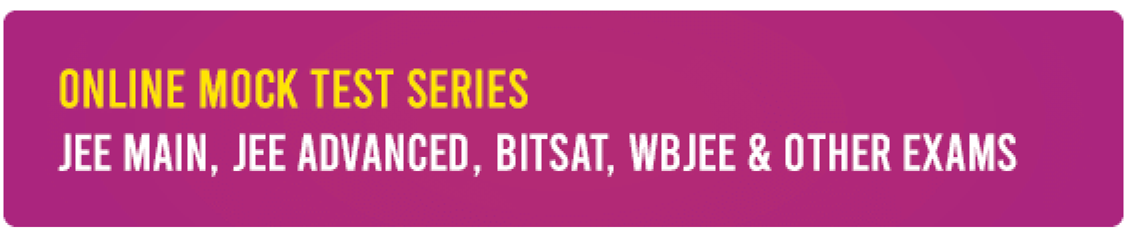

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JEE Mains 2020 Jan Chapter wise Question Bank

Properties of Matter

Q1

A string of length 60 cm, mass 6gm and area of cross section $1mm^2$ and velocity of wave 90m/s. Given young's modulus is $Y = 1.6 \times 10^{11}$ N/m^2. Find extension in string

एक 60 cm लम्बी रस्सी, जिसका द्रव्यमान 6gm और अनुप्रस्थ काट का क्षेत्रफल $1mm^2$ और तरंग का वेग 90m/s है। (दिया है यंग प्रत्यास्थता गुणांक $Y = 1.6 \times 10^{11}$ N/m^2) रस्सी में प्रसार बताइए

(1) 0.3 mm (2) 0.2 mm (3) 0.1 mm (4) 0.4 mm

7th Jan Morning

Sol

(1)

$$v = \sqrt{\frac{T}{\mu}}$$

$$T = \mu v^2$$

$$\frac{\mu v^2}{A} = Y\frac{\Delta \ell}{\ell}$$

$$\Delta\ell = \frac{\mu v^2 \ell}{AY}$$

after substituting value of μ, v, ℓ, A and Y we get

μ, v, ℓ, A तथा Y का मान रखने पर

$\Delta\ell = 0.3$ mm

Q2

A non-isotropic solid metal cube has coefficients of linear expansion as 5×10^{-5} /°C along the x-axis and 5×10^{-6} /°C along y-axis and z-axis. If coefficient of volume expansion of the solid is $C \times 10^{-6}$ /°C then the value of C is

7th Jan Morning

Sol

60

$V = 2\alpha_2 + \alpha_1$

$= 10 \times 10^{-6} + 5 \times 10^{-5}$

$= 60 \times 10^{-6}$ /°C

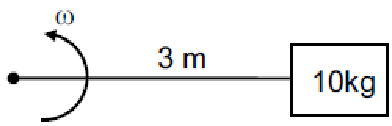
Q3

A wire of length ℓ = 3m and area of cross section $10^{-2}cm^2$ and breaking stress $48\times10^7 N/m^2$ is attached with block of mass 10kg. Find the maximum possible value of angular velocity with which block can be moved in circle with string fixed at one end.

9^{th} Jan Morning

Sol

4 rad/s



$\frac{T}{A} = \sigma$ ....(1)

$T = m\omega^2\ell$ ....(2)

Solving

ω = 4 rad/s

Q4

Ratio of energy density of two steel rods is 1 : 4 when same mass is suspended from the rods. If length of both rods is same then ratio of diameter of rods will be.

दो स्टील की छड़ों में ऊर्जा घनत्व का अनुपात 1 : 4 है, जब इनसे समान द्रव्यमान लटकाया जाता है। यदि इनकी लम्बाईयाँ समान है तो इनके व्यासों का अनुपात ज्ञात करो–

(1) $\sqrt{2} : 1$ (2) $1 : \sqrt{2}$ (3) 1 : 2 (4) 2 : 1

9^{th} Jan Evening

Sol

(1)

$$\frac{du}{dv} = \frac{1}{2}\text{ stress(प्रतिबल)} \times \frac{\text{stress}}{y}$$

$$= \frac{1}{2}\frac{F^2}{A^2 y}$$

$$\frac{du}{dv} \propto \frac{1}{d^4}$$

$$\frac{\left(\frac{du}{dv}\right)_1}{\left(\frac{du}{dv}\right)_2} = \frac{d_2^4}{d_1^4} = \frac{1}{4}$$

$$\frac{d_1}{d_2} = (4)^{1/4}$$

$$\frac{d_1}{d_2} = \sqrt{2} : 1$$